

Splendor DRL with Event Based Reward Shaping and Double DQN

FRA503 Deep Reinforcement Learning

Presented by

Chantouch (6611) and Sasish (6676)



Goal

- Win the game by reaching **at least 15 prestige points faster than the opponent**, using the fewest turns and the fewest cards possible to break any ties.



Problem Statements

Sparse Reward { The agent receives **no immediate feedback** for most actions, such as gem acquisition. Since Prestige Points are only awarded after a complex sequence of moves.

Delayed Strategic Payoff { Decisions regarding gem collection and card purchases **compound over many turns** before affecting the final prestige score.

State-Dependent Legal Action Space { The valid **action set changes every step** based on gem counts and card visibility. This requires a binary legal mask to be recomputed at every step.

Challenges

1. Reward Designing - **Points are too rare** to find by accident, but **too many small rewards make the agent distract** from winning
2. Legal Action Management - No rulebook information, **wasting time on the illegal moves.**
3. Temporal Strategic Depth - The math naturally **calculate getting a gem now over a discount** that pays off later.
4. State Complexity - **The board change every turn**, which is difficult to recognize pattern or strategic.

Rinascimento (Bravi & Lucas): using event-value functions for playing Splendor

- Theoretical Concept of Event-value Function (EF) and Implementation

$$h_w(E_{s \rightarrow s'}) = f_w(\sigma(E_{s \rightarrow s'})) \longrightarrow r'_t = r_t^{score} + \phi_t^\top \mathbf{w}$$

Event	Description	Trigger	Weight w_i
ϕ_1	Collect gem(s)	Gem count increases this step	+0.05
ϕ_2	Purchase a card	A card is acquired this step	+0.20
ϕ_3	Gain a card bonus	Any bonus-color count increases	+0.10
ϕ_4	Attract a noble	A noble is attracted this step	+0.50
ϕ_5	Win the game	Agent prestige ≥ 15 at turn end	+1.00

Baseline reward shaping and weights

Value Estimation (van Hasselt et al.): DRL with Double Q-learning

- The Bellman Update Target Equation

$$y_t = r_t + \gamma Q(s_{t+1}, \arg \max_{a'} Q_{masked}(s_{t+1}, a'; \theta); \theta^-) \quad \text{For predict the future by using the discount factor}$$

+

Have a TARGET NETWORK for prevent runaway feedback loops.

Legal Action Masking (Huang & Ontañón): A Closer Look at Invalid Action Masking in Policy Gradient Algorithms

- Masking Technique

$$Q_{masked}(s, a; \theta) = \begin{cases} Q(s, a; \theta), & m_a = 1 \\ -\infty, & m_a = 0 \end{cases}$$

To prevent the network from attempting invalid moves
by implementing the cover vector [1, 0, 1, ..., 1]

ε-Greedy Exploration with Exponential Decay (Sutton & Barto): RL Introduction

- Exploration Strategy

$$\epsilon_k = \max(\epsilon_{min}, \epsilon_0 \cdot \lambda^k)$$

Using decaying rate

Experience Replay (V. Mnih et al.): Human-level control through DRL

- Mean Squared Error

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{B}}[(y_t - Q(s, a; \theta))^2]$$

Using MSE as a loss for ensures negative and positive errors don't cancel each other out, and it mathematically punishes the network harder for making large mistakes.

Objective

1. **Build a Splendor game** environment that works with DRL frameworks
2. **Implement a DDQN agent** with action masking and experience replay
3. Design an **event-based reward shaping** scheme
4. **Build a visualization** that supports both AI vs AI and AI vs Human modes
5. **Evaluate the agent's** though metrics

Scope

- **2-player** Splendor only
- Core game mechanics
- Two play modes (AI vs AI for training, AI vs Human)
- **Comparison** between two reward conditions only (score-only vs event-shaped)
- **Analyze the agent behavior** in Splendor
- **No guarantee the agent beats a strong human player**

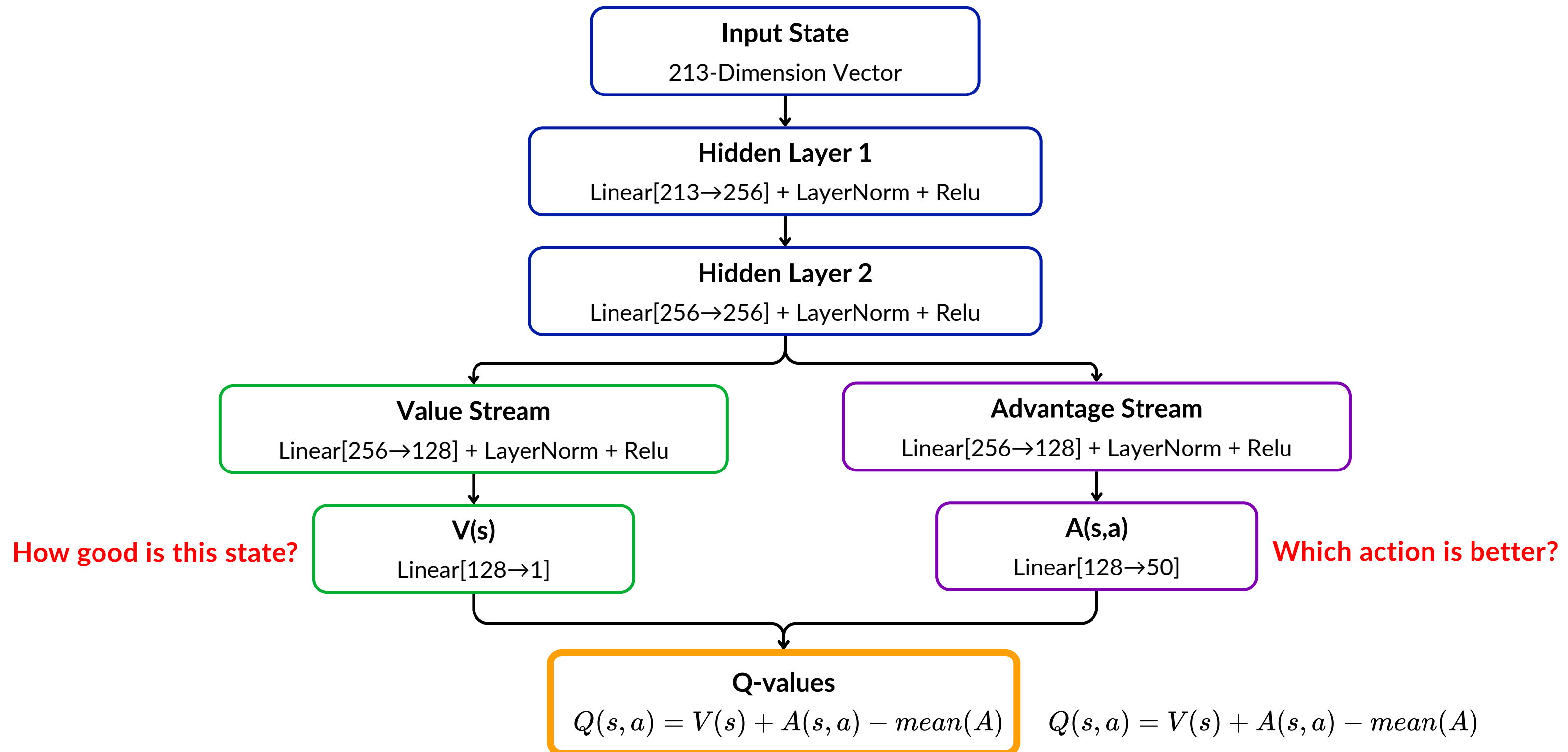
State Encoder

Component	Size
Face-up cards on board (3 rows × 4 cards)	132
Noble tiles	18
Board gem supply	6
Agent state (gems, bonuses, prestige, reserved)	45
Opponent state (gems, bonuses, prestige)	12
Total	213

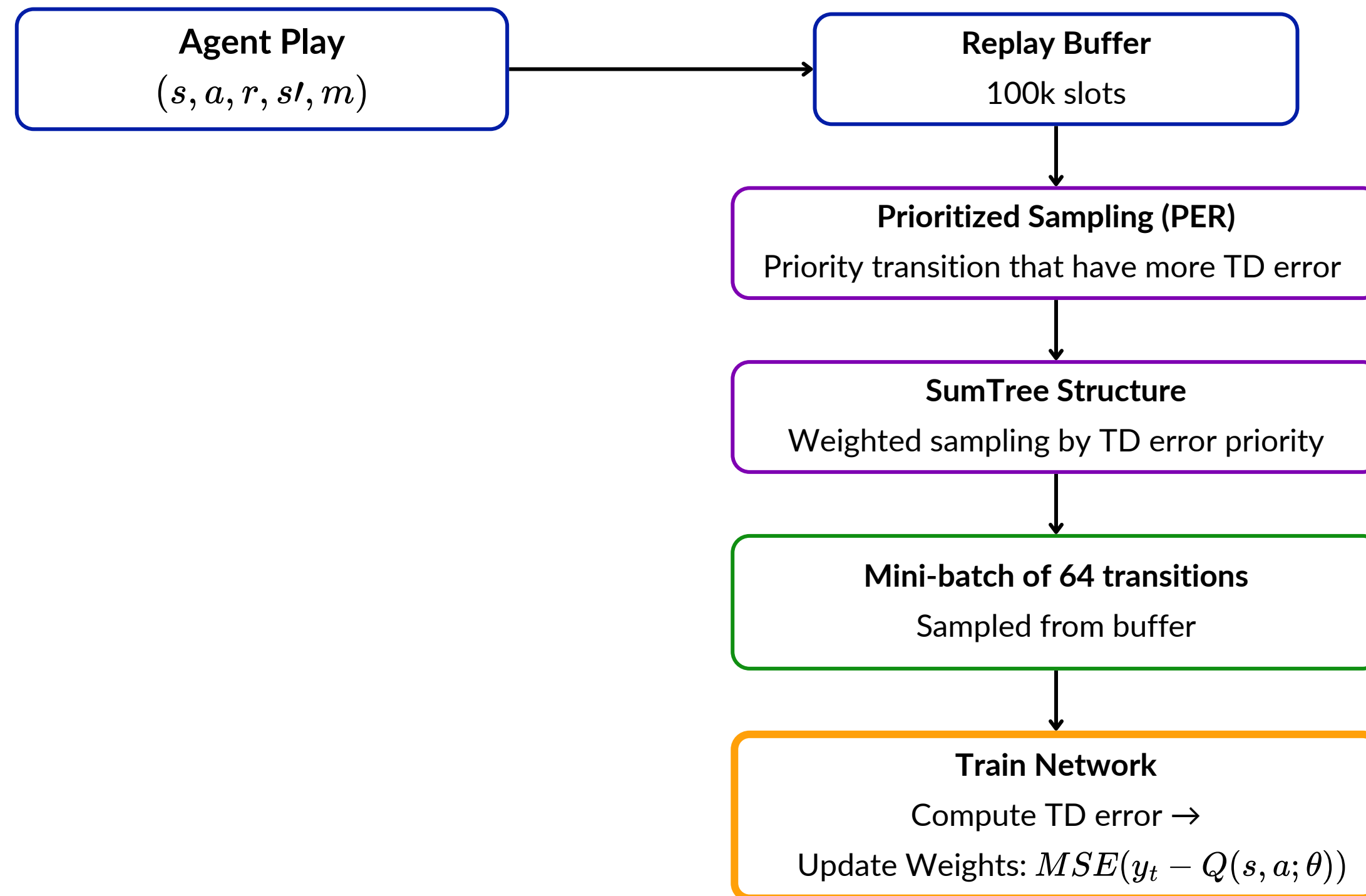
Action Space

Action	Explanation	Count
Take gems	10 three-different-color combos + 5 two-same-color	15
Buy Card	12 face-up cards + 3 reserved cards in hand	15
Reserve Card	12 face-up cards + 3 blind draws from deck	15
Discard Gem	Return 1 gem per color (only when tokens > 10)	5
Total		50

Dueling Q-Network Architecture



Experience Replay



TD Error = Actual - Expects

Legal Action Masking

- 50-action array, each slot True or False (m)
- Generated fresh every turn by the environment
- Based on current board state (enough gems? can afford card? hand full?)

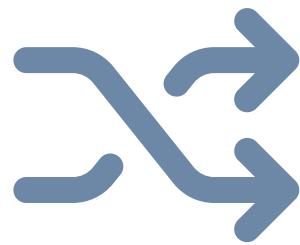
DDQN Output					
Q-values	2.1	0.8	-0.3	1.5	0.2
Mask	✓	✗	✓	✓	✗
Masked action	2.1	$-\infty$	-0.3	1.5	$-\infty$
argmax	2.1				

Applied at action selection and during training

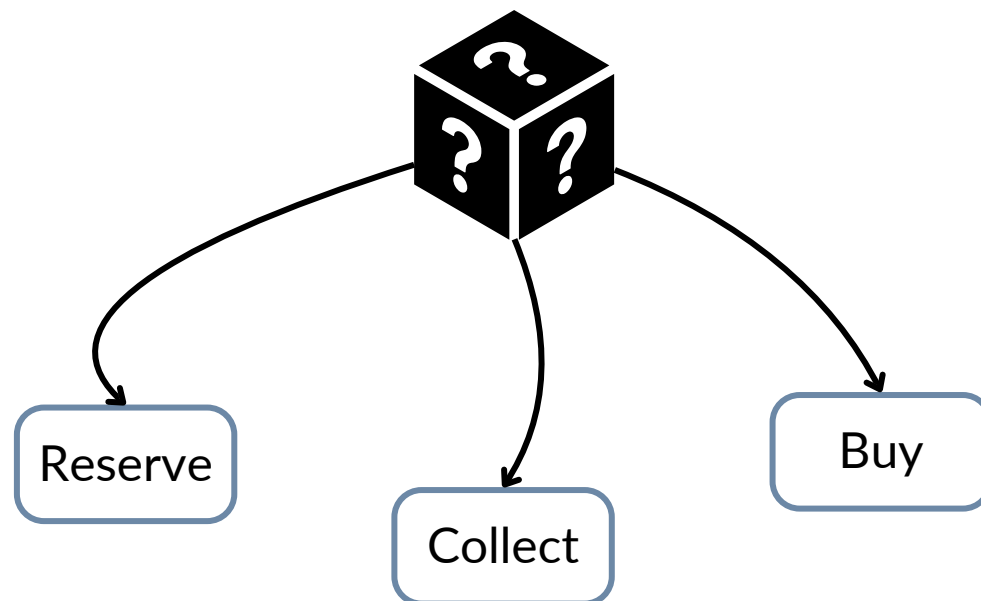
Technical Implementation

Opponent

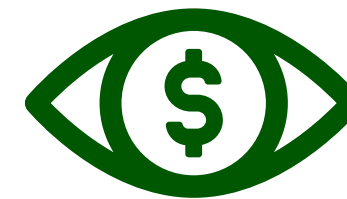
Random



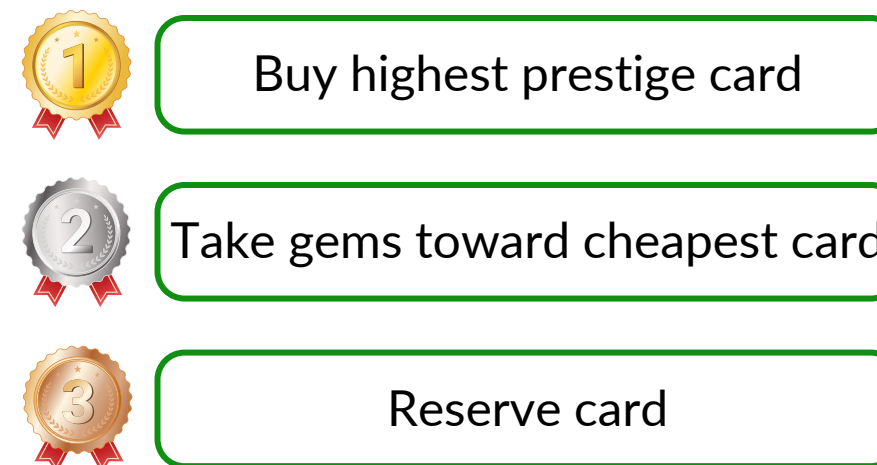
- Picks any legal action with equal probability



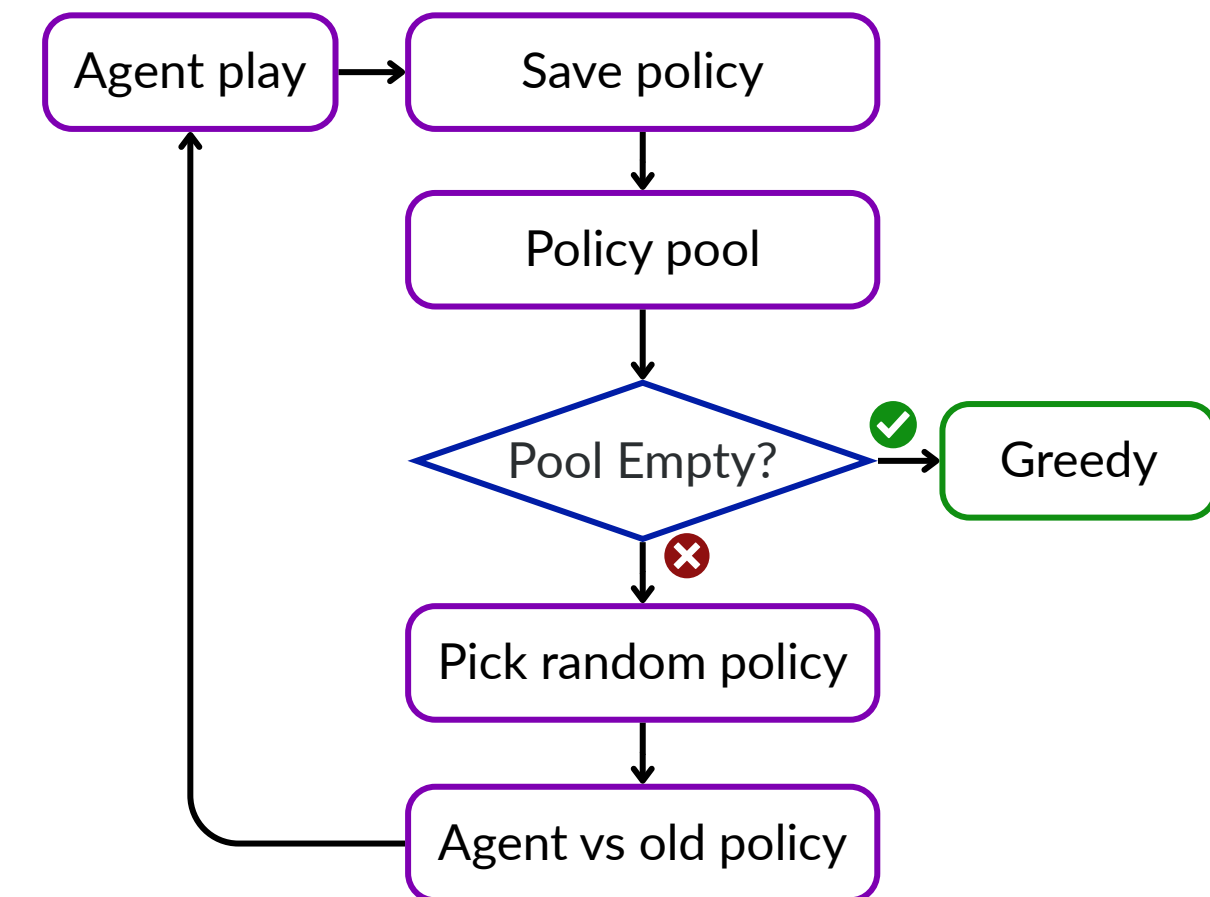
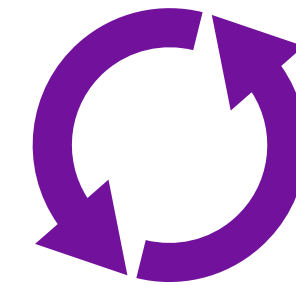
Greedy



- Fixed priority list:



Self-play



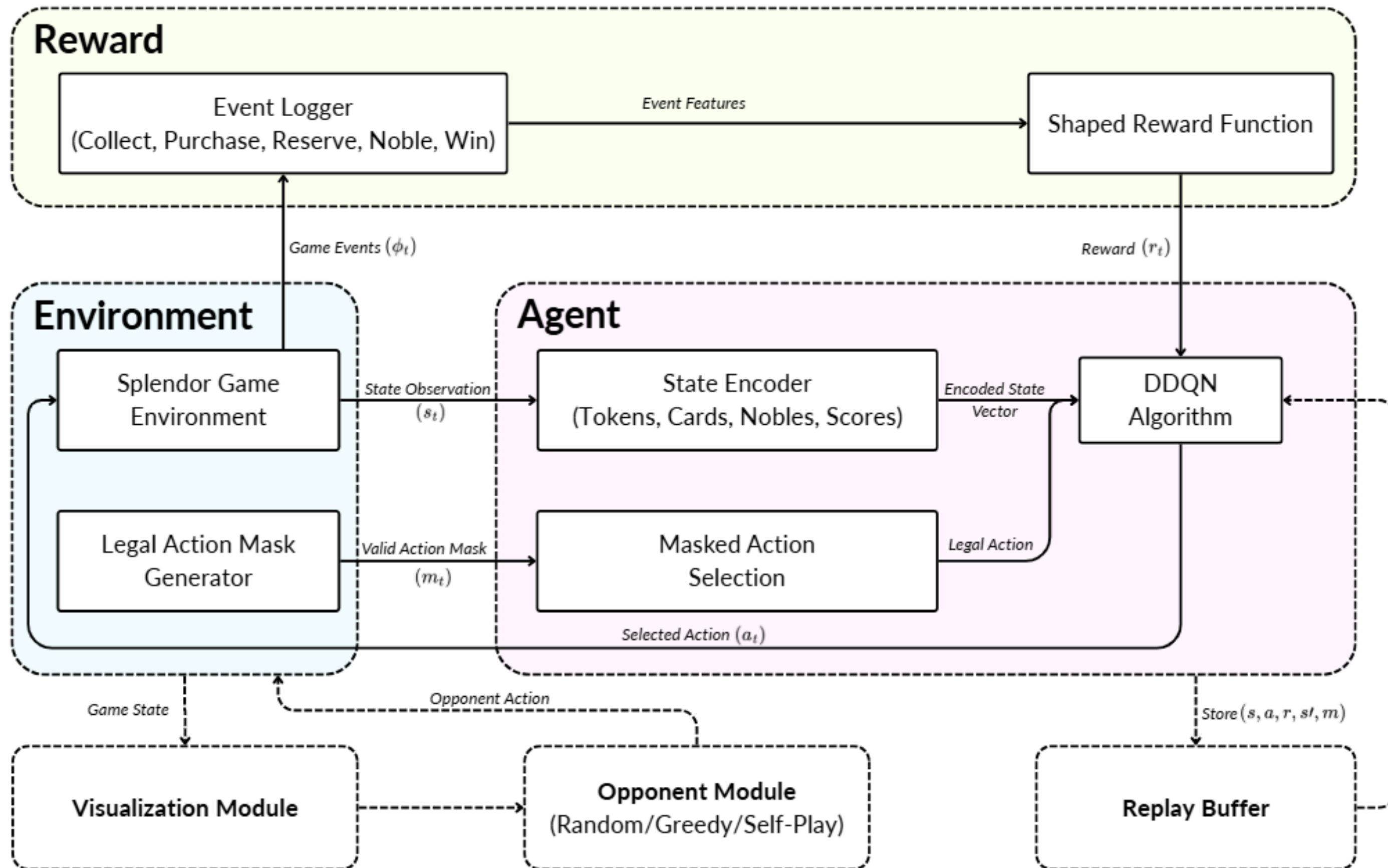
Event-based Reward

Event	Description	C2 (Baseline)	C4 (Card Rush)	C5 (Noble Hunter)	C6 (Balanced Dense)
Gem Taken	Token collected from board	+0.002	+0.002	+0.002	+0.010
Card Bought L1	Level-1 development card purchased	+0.05	+0.05	+0.05	+0.10
Card Bought L2	Level-2 development card purchased	+0.10	+0.10	+0.10	+0.20
Card Bought L3	Level-3 development card purchased	+0.20	+0.20	+0.20	+0.40
Prestige Gained	Victory points scored	+1.00	+1.00	+1.00	+1.00
Noble Gained	Noble tile visited	+3.00	+0.50	+6.00	+4.00
Card Reserved	Card held in hand	+0.01	+0.01	+0.01	+0.02
Win Bonus	Game won	+10.00	+10.00	+10.00	+10.00
Loss Penalty	Game lost	-10.00	-10.00	-10.00	-10.00

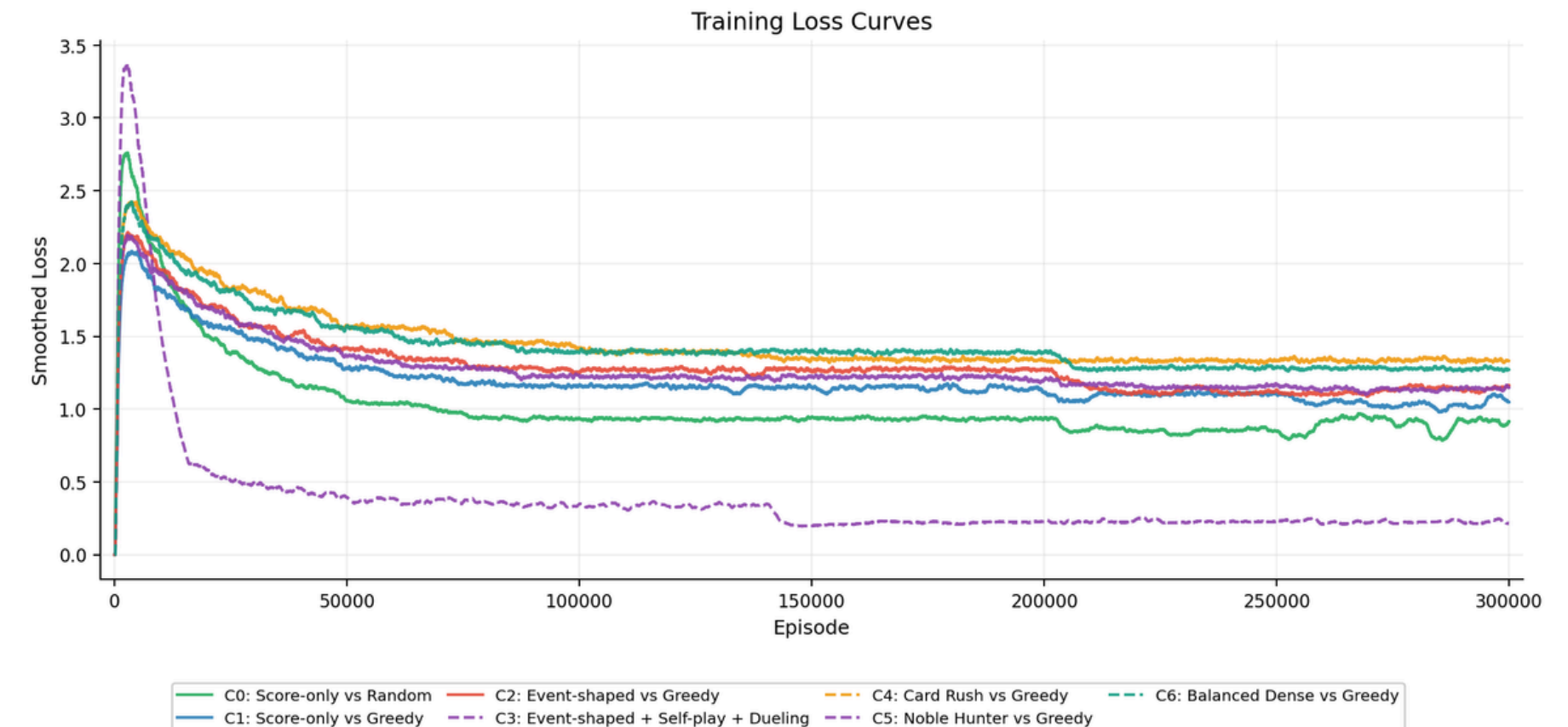
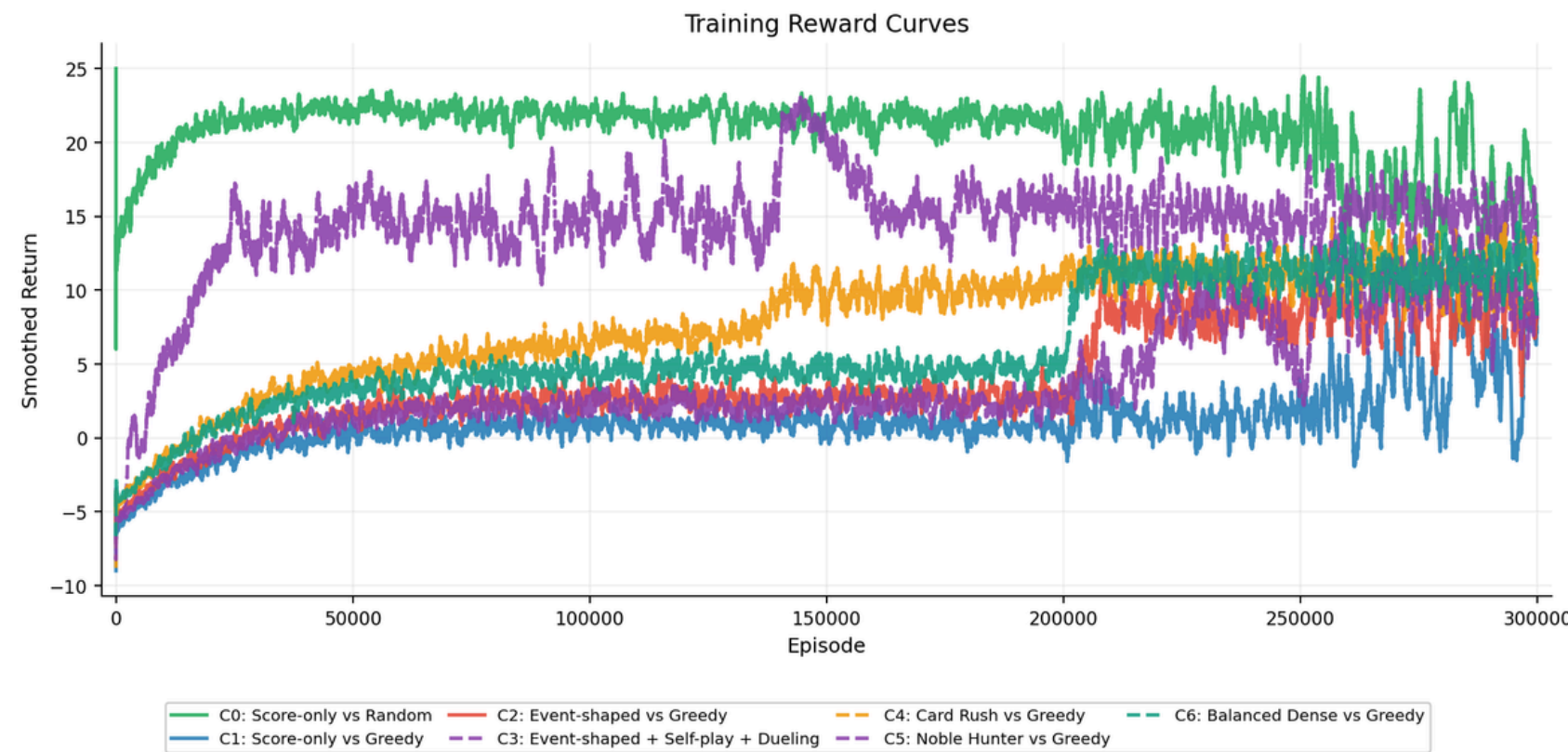
Experimental Setup

Condition	Reward	Opponent	What it tests
C0	Score-only	Random	Lower bound
C1	Score-only	Greedy	Baseline
C2	Event-shaped (base)	Greedy	Value of reward shaping
C3	Event-shaped + Self-play	Self	Value of opponent curriculum
C4	Event-shaped + Card Rush	Greedy	Does buying cards faster help?
C5	Event-shaped + Noble Hunter	Greedy	Does targeting nobles improve performance?
C6	Event-shaped + Balanced Dense	Greedy	Does denser reward signal accelerate learning?

System Overview



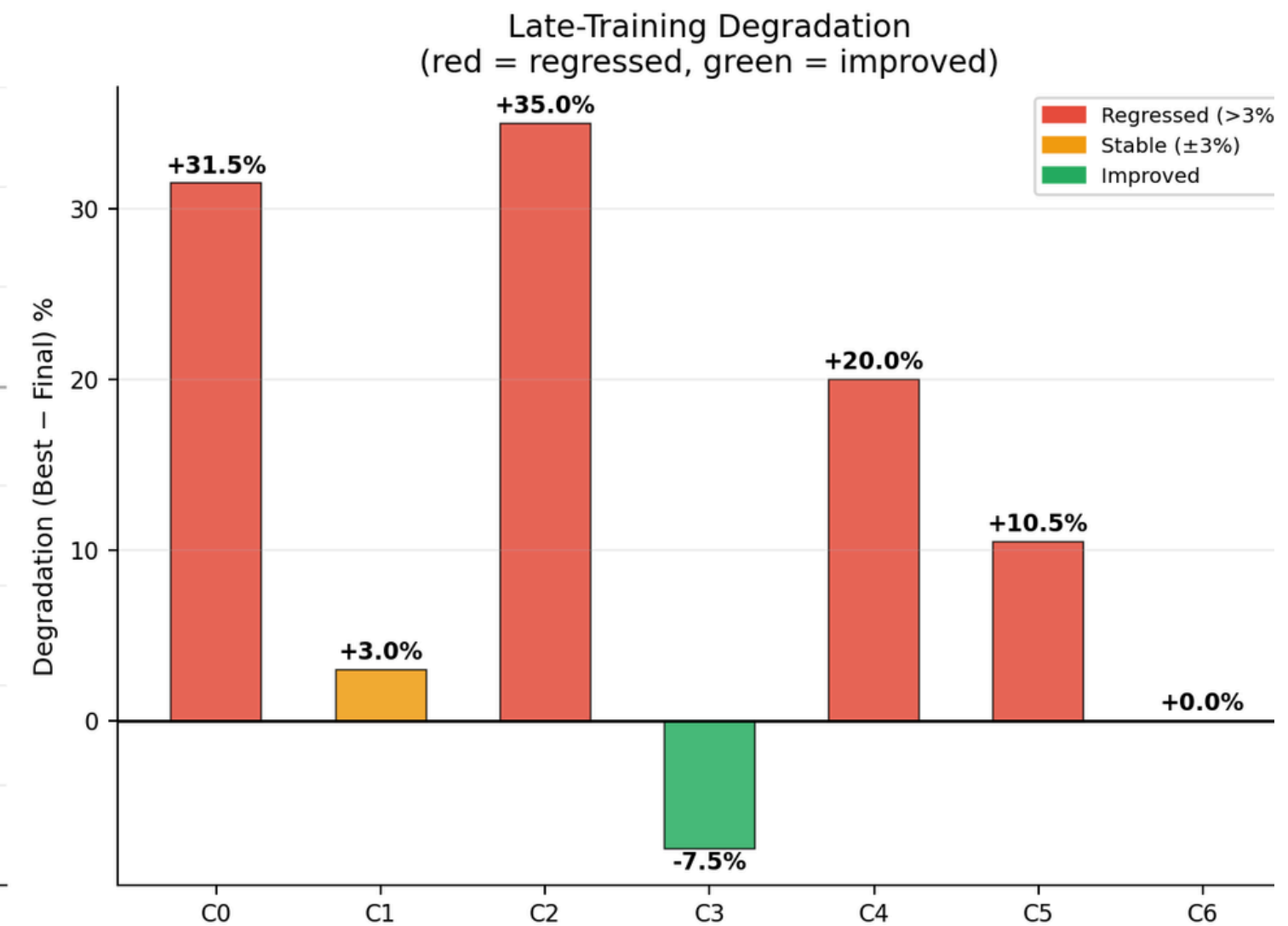
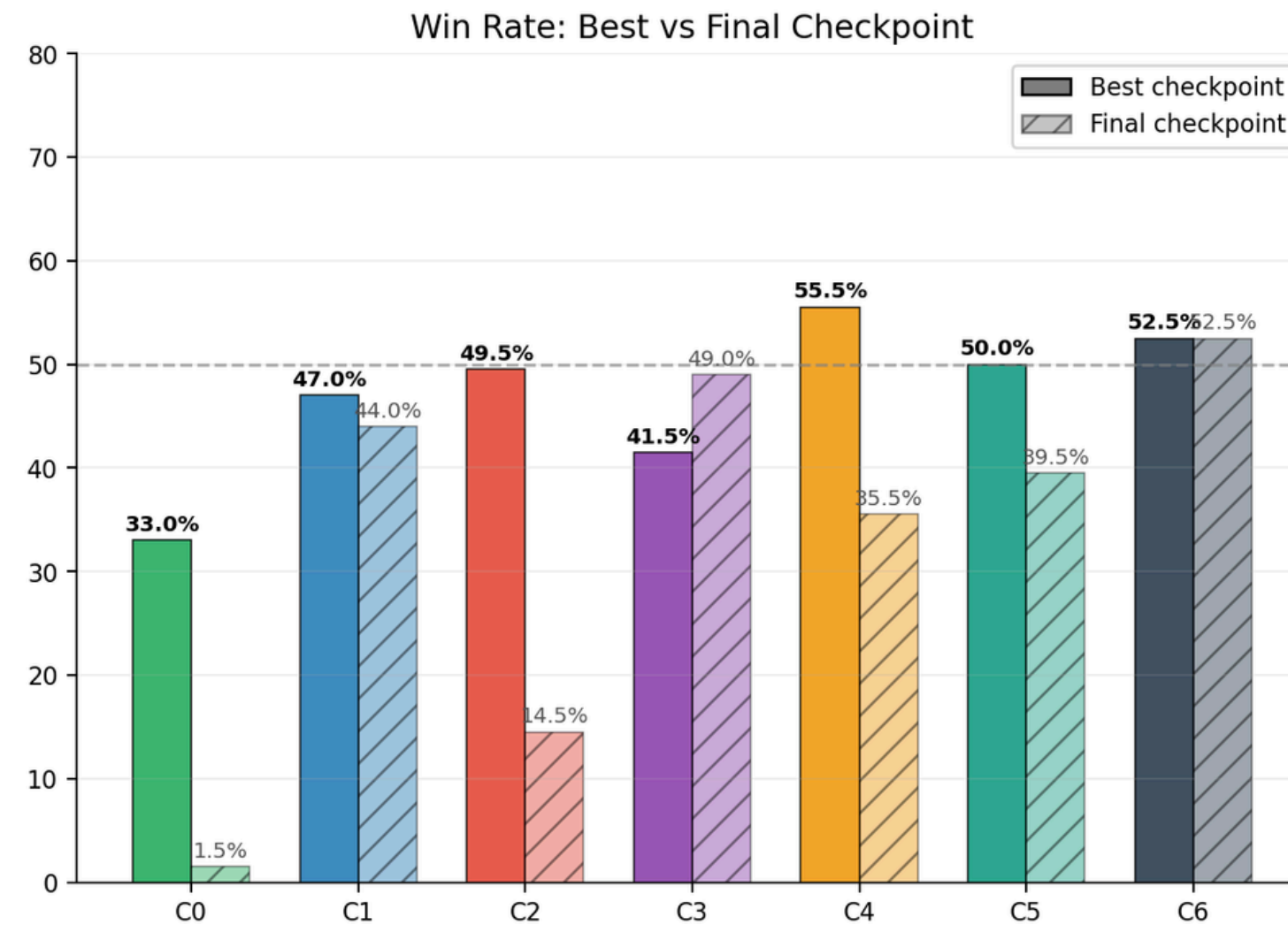
Training Reward & Loss



Event-based reward shaping **successfully** mitigates the Sparse Reward problem.

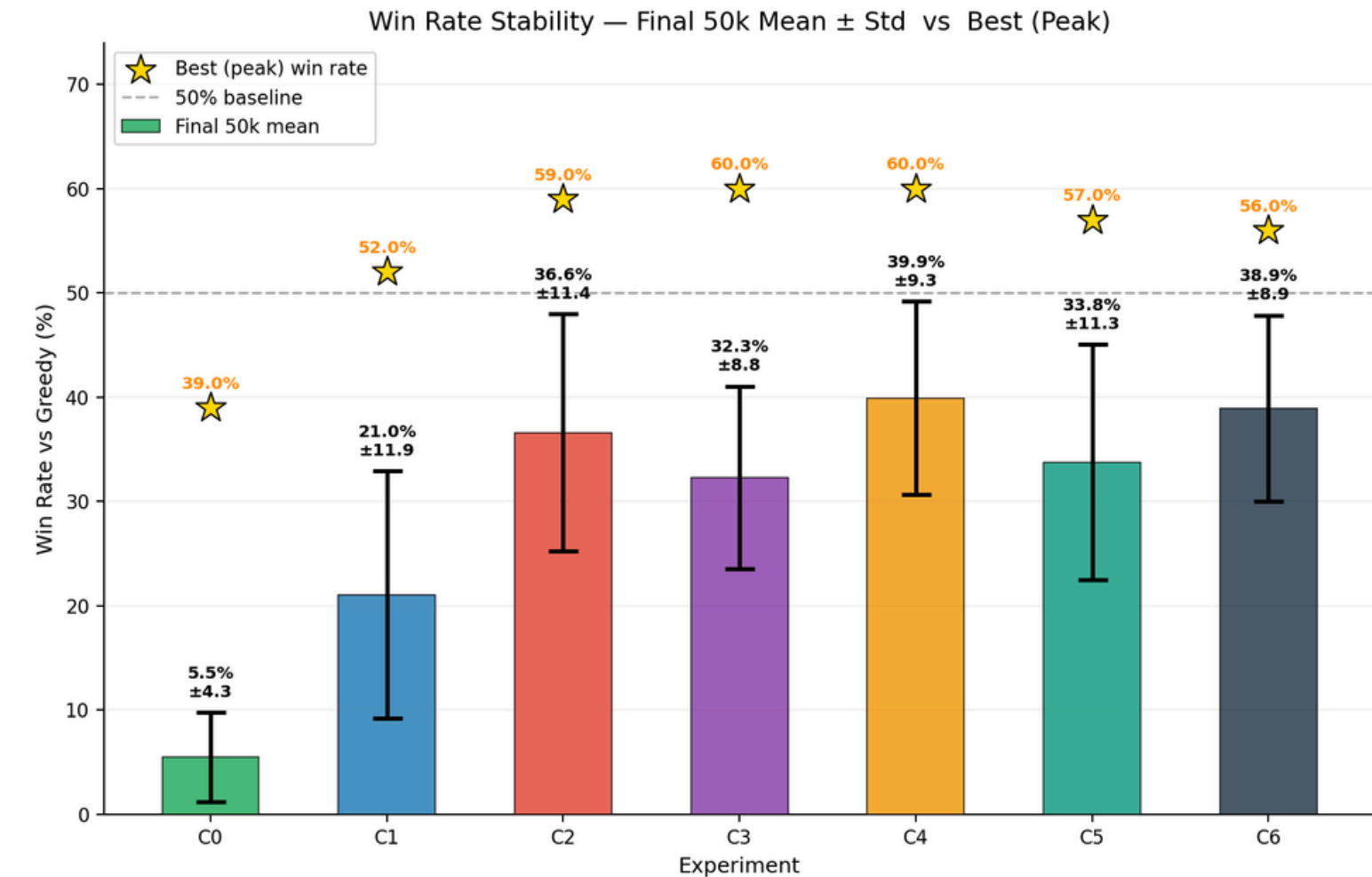
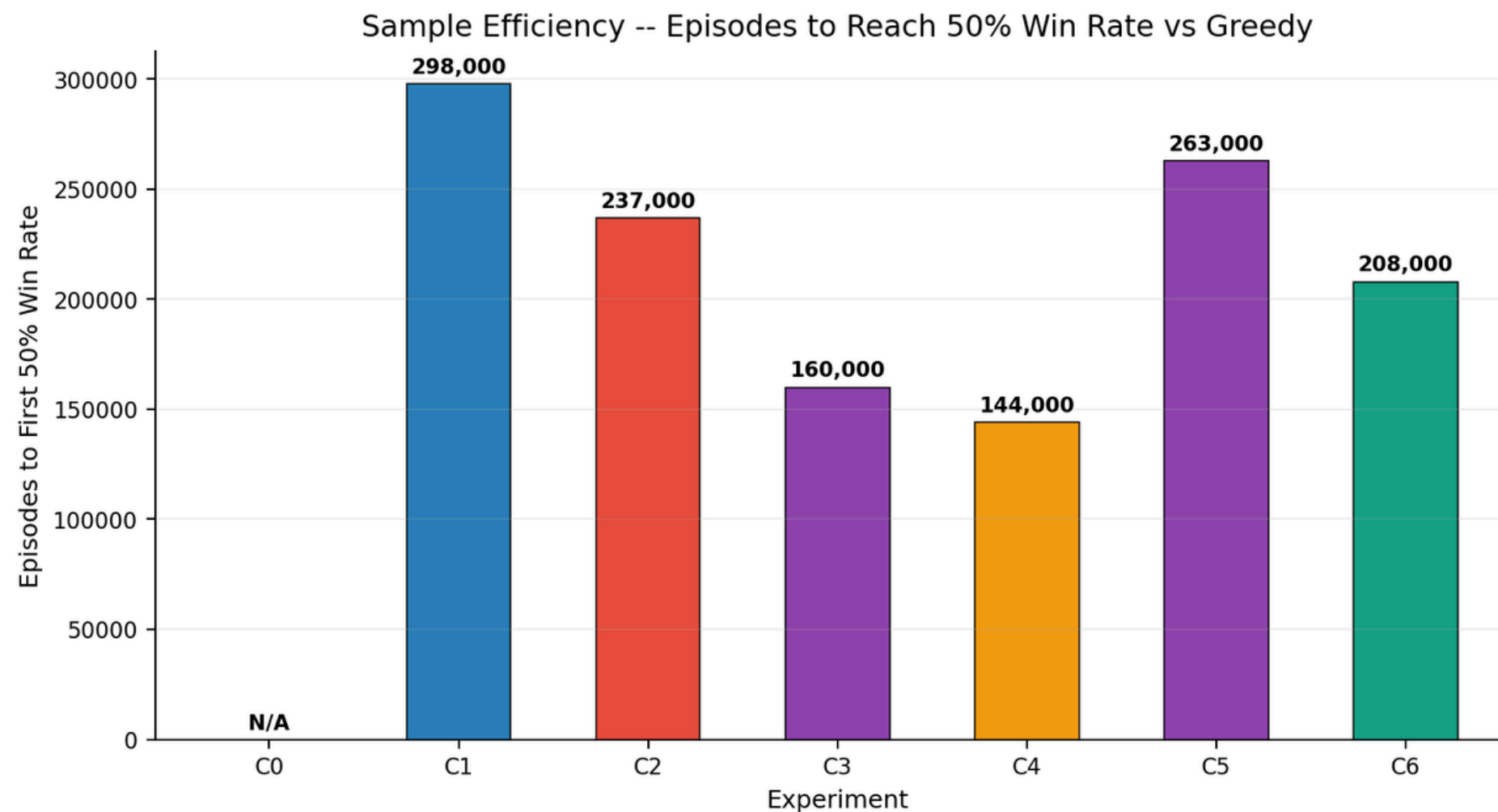
C3 has a spike at ~150,000 episodes from **Self-play breakthrough** (agent dominating its pool) immediately. And the noise after 200,000 episodes reflect the **epsilon decreasing** $1.0 \times (0.99997)^n = 0.05 \rightarrow n \approx 99,857$ episodes which makes the static policy and the nature of the Splendor mechanics is change the action space every step this shows the agents are **not generalize**.

Did training too long hurt?



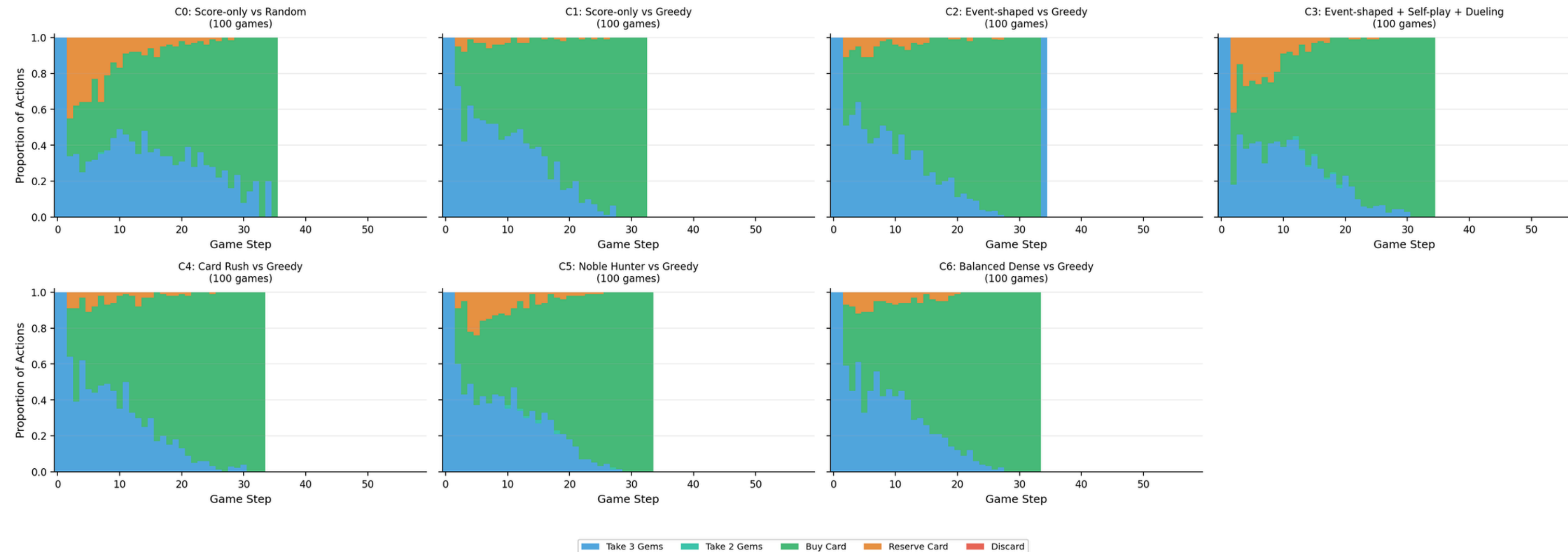
The C3 is the only one that have **better average final checkpoint** which represent the intelligence of the agent that have self-play which cause the breakthrough

Result



C4 learns fastest reaches 50% win rate in just **144,000 episodes**, nearly half of what C1 needs (298,000). C3 is second at 160,000 despite its heavier architecture. C5 (Noble Hunter) is surprisingly slow at 263,000 — the noble strategy requires a long multi-step chain that is hard to credit early in training. **C0 never reaches 50%** training vs random **doesn't generalize to greedy**.

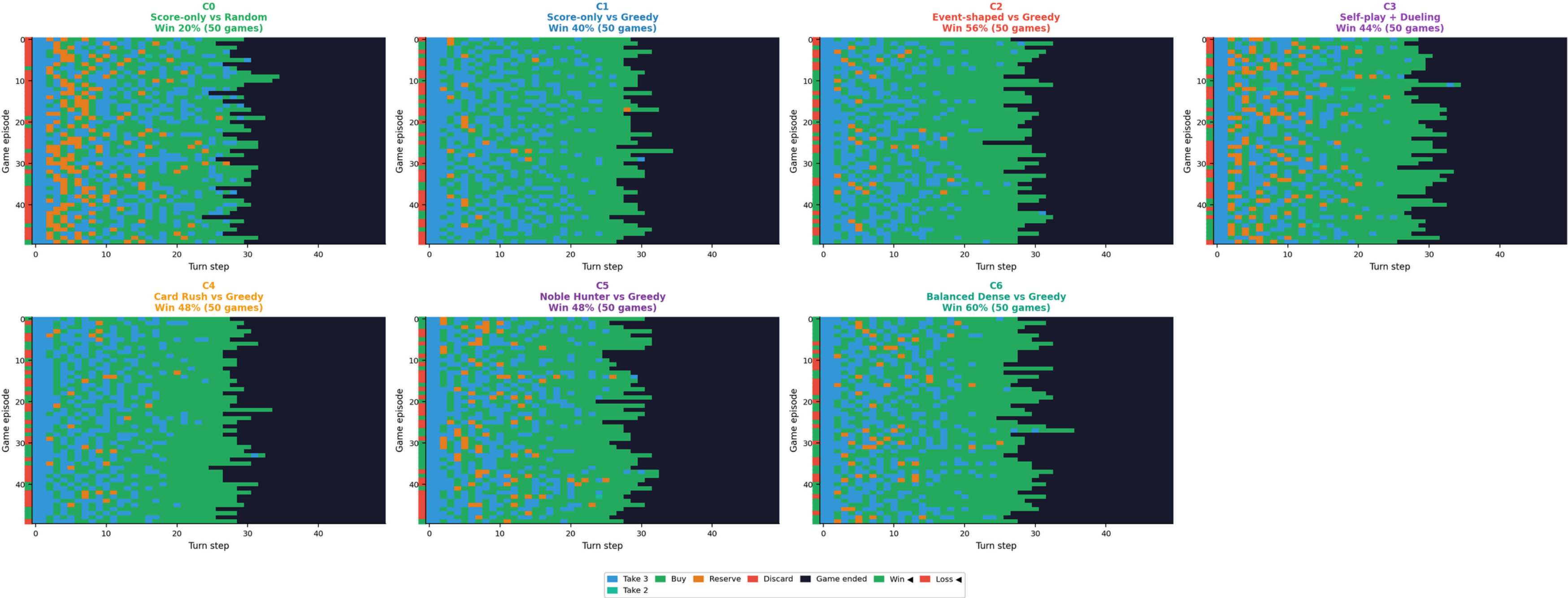
Sequential Action Distribution by Game Step



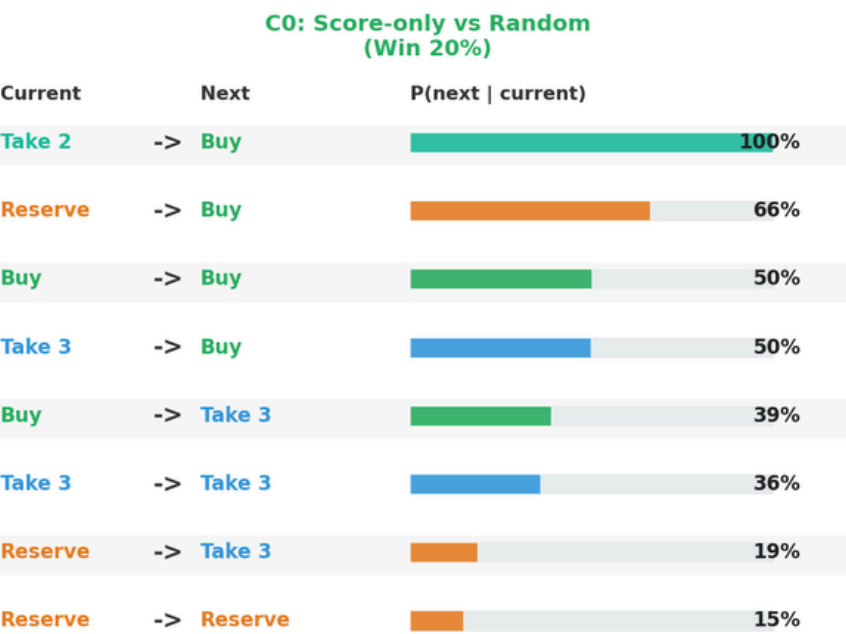
Every agent follows the same **two-phase structure**:

- **Early game (steps 0–15):** Dominated by **blue (Take 3 Gems)** — agents must collect resources before they can spend them
- **Late game (steps 15+):** **Green (Buy Card)** grows and eventually dominates — gems have been accumulated, now the agent converts them into cards. This is actually correct Splendor strategy — it mirrors how human players behave (build economy first, spend later).

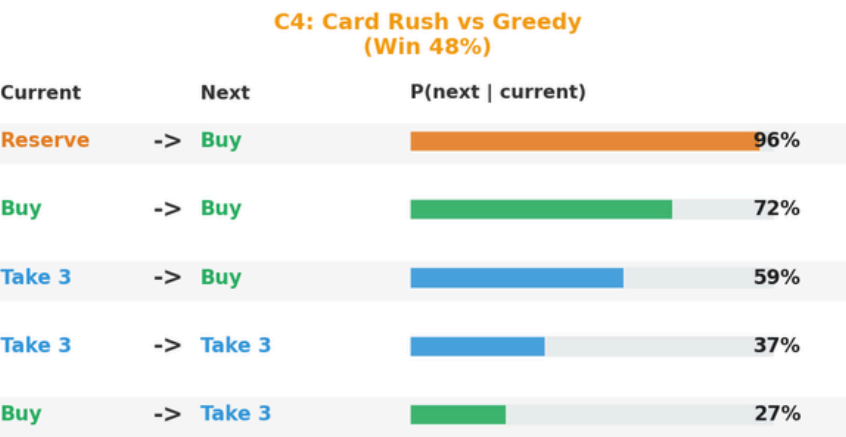
Action Ethogram



Action Transitions



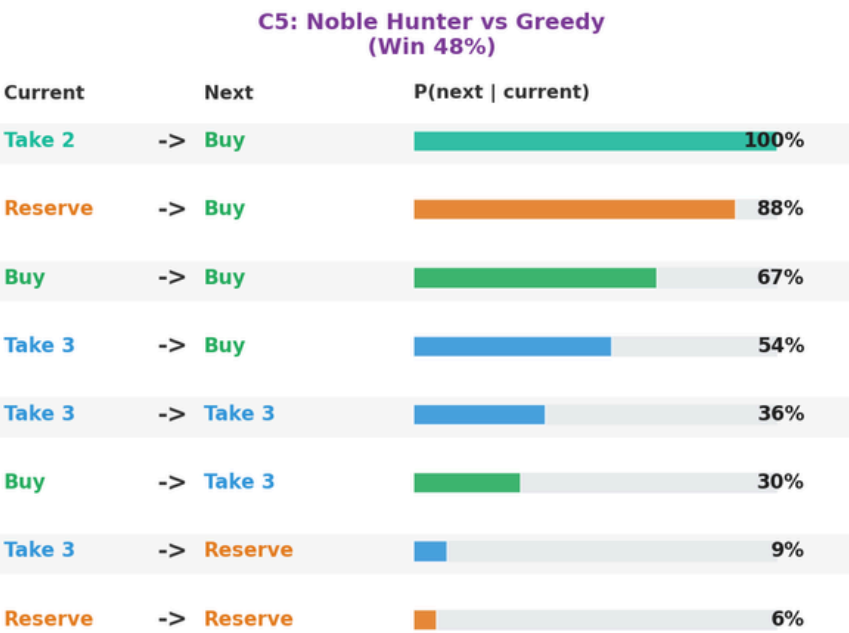
Start mix: Buy 49%, Take 3 38%, Reserve 12%



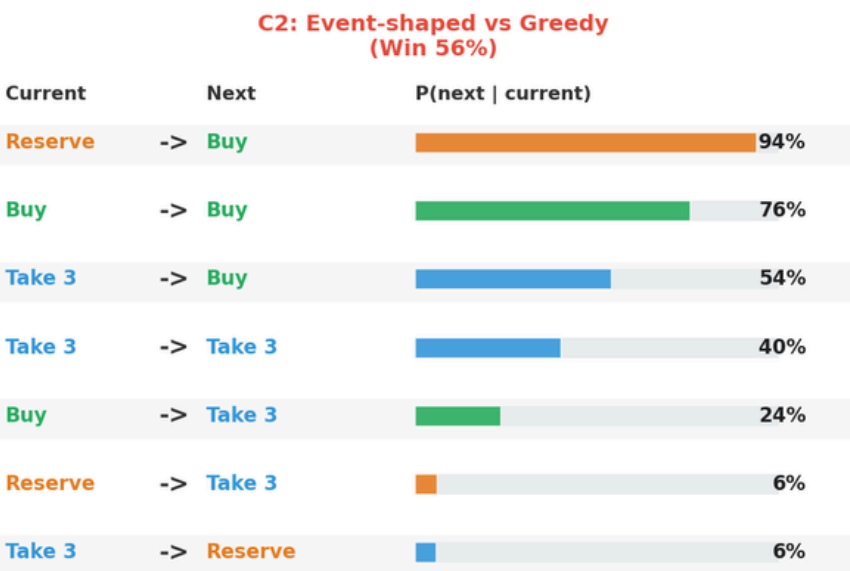
Start mix: Buy 65%, Take 3 33%, Reserve 2%



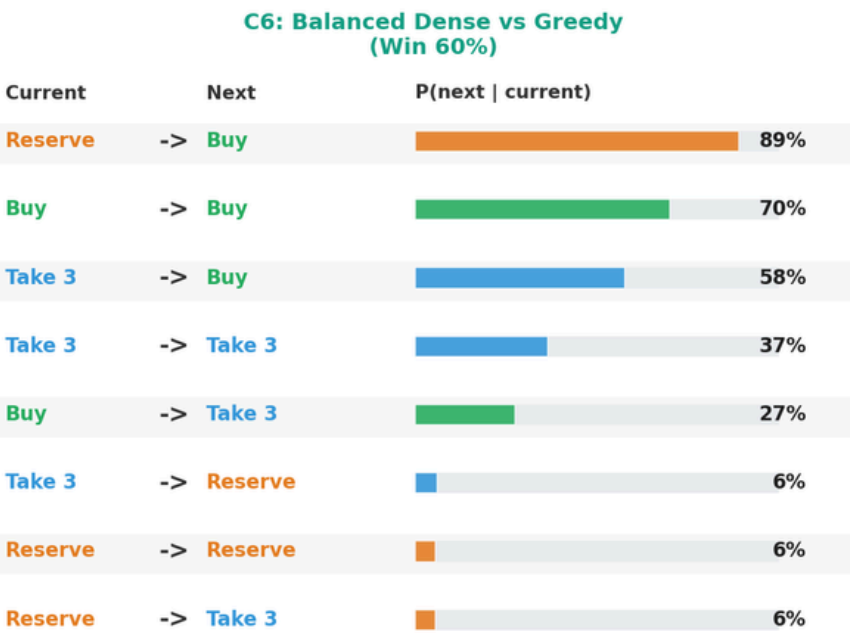
Start mix: Buy 61%, Take 3 37%, Reserve 2%



Start mix: Buy 60%, Take 3 34%, Reserve 6%



Start mix: Buy 66%, Take 3 32%, Reserve 2%



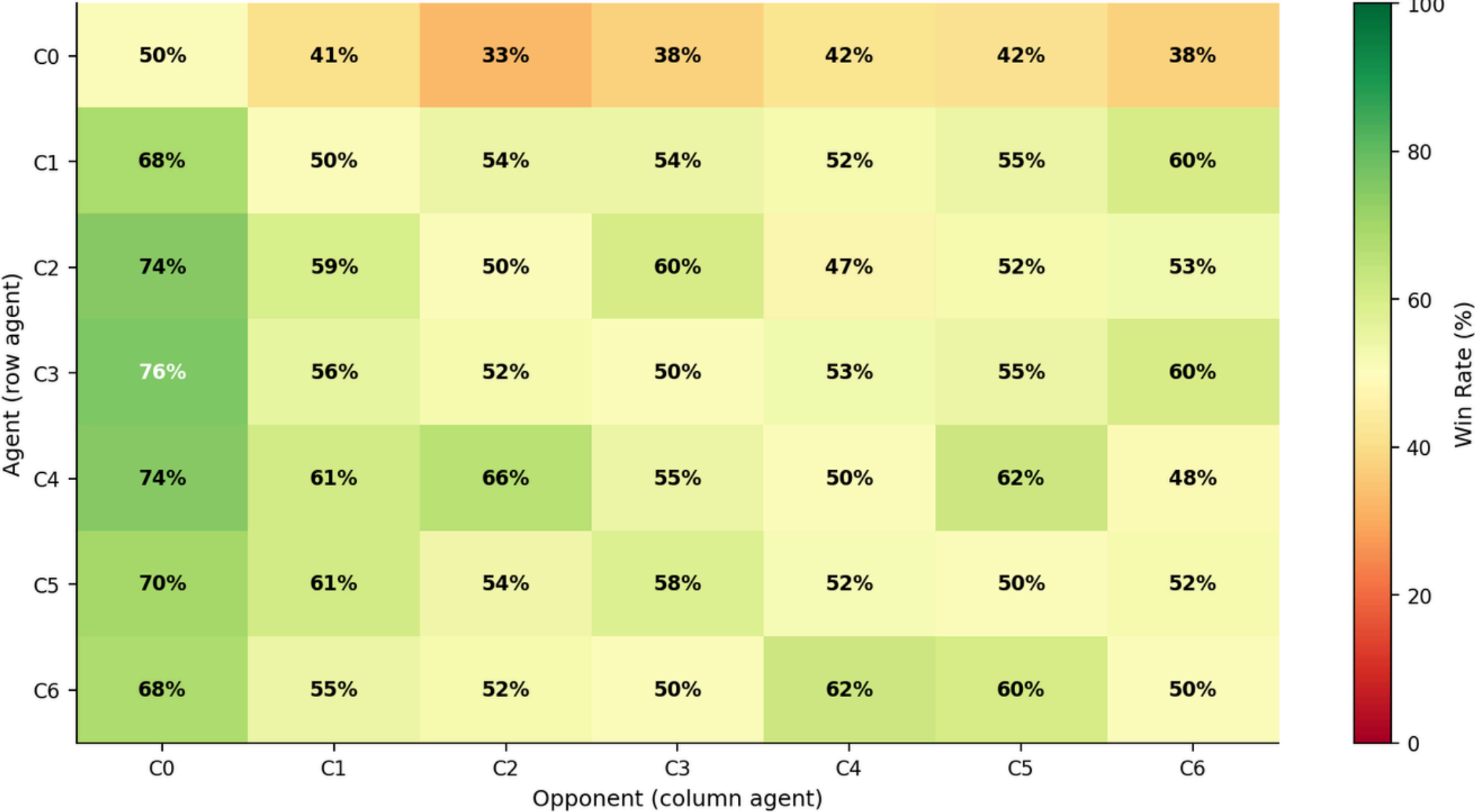
Start mix: Buy 63%, Take 3 33%, Reserve 4%



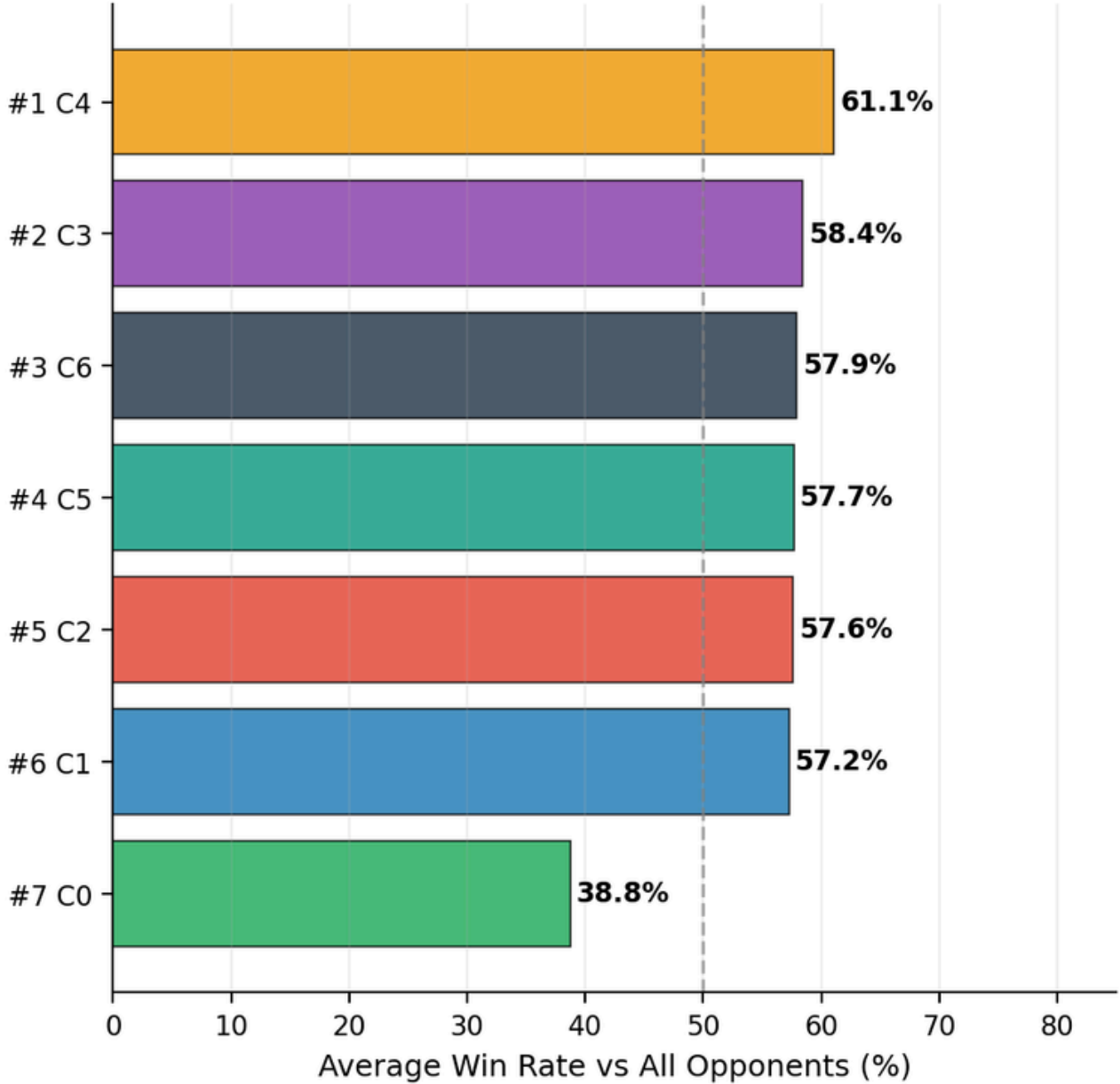
Start mix: Buy 61%, Take 3 30%, Reserve 8%

Cross-Experiment Tournament

Win Rate Matrix
(row = player, col = opponent)



Tournament Ranking
(avg win rate vs all others)



Problem Statement /Challenges	Solved?	Evidence
Sparse Reward	Yes	Event-based shaping (C2–C6) gives immediate feedback per action — reward curves rise significantly above C0/C1 baseline
Delayed Strategic Payoff	Partially	C4 (card rush) and C5 (noble hunter) redirect agent toward long-term goals, but full convergence not achieved within 300k episodes
State-Dependent Legal Action Space	Yes	Legal mask eliminates illegal moves completely — no red (discard) in most agents, zero wasted illegal actions
Reward Designing	Partially	C4/C5/C6 ablations show reward structure matters — but the right balance is still not found; all final-phase means below 50%
Temporal Strategic Depth	Partially	Discount Factor = 0.99 + boosted future rewards help, but agents still prefer short-term payoffs as seen in volatile late-game behavior
State Complexity	Partially	Dueling DQN + LayerNorm helps (C3 has lowest loss), but high irreducible variance (std ~14) shows the board state is still too complex to master

 สถาบันวิทยาการหุ่นยนต์ภาคสนาม (FIBO)
 Institute of field Robotics

SPLENDOR
Deep Reinforcement Learning

Turn 4 | AI vs Human | Speed: 1.0s +/- adjust ESC quit R restart

Level 3 (16 remaining)

4 +Dia
7 Ony

4 +Rub
7 Eme
3 Rub

4 +Ony
7 Rub
3 Ony

4 +Dia
3 Dia
7 Ony

Level 2 (26 remaining)

1 +Rub
3 Sap
2 Rub
3 Ony

2 +Dia
5 Rub

1 +Sap
2 Sap
3 Eme
3 Ony

2 +Sap
5 Sap

Level 1 (34 remaining)

+Dia
4 Eme

+Rub
2 Sap
1 Eme

+Eme
2 Sap
2 Rub

+Rub
1 Dia
1 Rub
3 Ony

Nobles

3 pts
4 4

3 pts
4 4

3 pts
4 4

Move History

T	Who	Action
2	You	Take 3 gems: green, red, black
2	AI	RES Reserve Level-1 card #4
4	You	Take 3 gems: white, blue, green
4	AI	RES Reserve Level-1 card #4

You

Tokens

1 Dia

1 Sap

2 Eme

1 Rub

1 Ony

0 Gld

Card Bonuses

0 Dia

0 Sap

0 Eme

0 Rub

0 Ony

Reserved (0/3)

AI Opponent

Tokens

0 Dia

0 Sap

0 Eme

0 Rub

0 Ony

2 Gld

Card Bonuses

0 Dia

0 Sap

0 Eme

0 Rub

0 Ony

Reserved (2/3)

Opt +Sap
3 3 4

Opt +Ony
1 1 1

Gem Supply (click to select)

3 Diamond

3 Sapphire

2 Emerald

3 Ruby

3 Onyx

3 Gold

Gem Selection — click board gems above, then Confirm

?

?

?

Select gems from the board above.

Confirm

Clear

Cards: top-half = BUY bottom-half = RESERVE

Visualization

SPLENDOR Deep Reinforcement Learning

Turn 0 | AI vs Human | Speed: 1.0s +/- adjust ESC quit R restart

Level 3 (18 remaining)

Level 2 (26 remaining)

Level 1 (36 remaining)

Gem Supply (click to select)

Diamond Sapphire Emerald Ruby Onyx Gold

Gem Selection — click board gems above, then Confirm

Select gems from the board above

Move History

No moves yet.

You 0

Tokens

Dia Sap Emc Rub Ony Gld

Card Bonuses

Dia Sap Emc Rub Ony

Reserved (0/3)

AI Opponent 0

Tokens

Dia Sap Emc Rub Ony Gld

Card Bonuses

Dia Sap Emc Rub Ony

Reserved (0/3)

Confirm Clear

Cards: top-half = BUY bottom-half = RESERVE

